

Asset Financing Restraint and the Cross Section of Stock Returns

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October 21, 2025

Abstract

This paper studies the asset pricing implications of Asset Financing Restraint (AFR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on AFR achieves an annualized gross (net) Sharpe ratio of 0.52 (0.40), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (16) bps/month with a t-statistic of 3.05 (2.32), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net debt financing, Net external financing, Change in financial liabilities, Asset growth, Inventory Growth, change in ppe and inv/assets) is 18 bps/month with a t-statistic of 2.57.

1 Introduction

Asset prices reflect investors' marginal rates of substitution between current and future consumption, linking cross-sectional return variation to firms' exposure to systematic consumption risk. While substantial progress has been made in understanding how firm characteristics relate to expected returns through the lens of consumption-based asset pricing models, the mechanisms through which corporate financing decisions affect firms' consumption risk exposure remain incompletely understood. We identify and examine Asset Financing Restraint (AFR) as a novel predictor of cross-sectional stock returns that captures firms' conservative approach to external financing relative to their asset base. This measure reflects managerial restraint in leveraging external capital markets to fund asset expansion, potentially signaling differential exposure to aggregate consumption risk during economic downturns when external financing becomes costly or unavailable.

The consumption-based asset pricing framework suggests that firms exhibiting high asset financing restraint earn higher expected returns because they bear greater exposure to systematic consumption risk (Cochrane et al., 2024; Bansal and Yaron, 2004). When firms maintain conservative external financing policies relative to their asset base, they become more sensitive to aggregate consumption shocks through multiple channels. First, financially constrained firms with limited external financing capacity face higher operating leverage, making their cash flows more procyclical and thus more correlated with the representative agent's marginal utility of consumption (Zhang, 2005; García-Feijóo and Jermann, 2005). During economic contractions when consumption growth is low and marginal utility is high, these firms experience disproportionate declines in profitability, requiring higher ex-ante returns to compensate investors.

The state-dependent nature of financing frictions amplifies this consumption risk exposure during bad economic times (He and Krishnamurthy, 2013; Brunnermeier

et al., 2016). Firms with high asset financing restraint maintain lower debt capacity buffers, leaving them vulnerable to tightening credit conditions that typically coincide with consumption disasters or near-disaster events. This vulnerability manifests as heightened sensitivity to long-run consumption risks, as temporary financing disruptions can permanently impair firms' productive capacity and future cash flow generation (Bansal et al., 2012). The inability to smooth investment through external financing forces these firms to bear undiversifiable technology shocks that covary with aggregate consumption, particularly during periods of high marginal utility.

Furthermore, asset financing restraint relates directly to the intertemporal marginal rate of substitution through its impact on firms' investment-consumption dynamics (Cochrane and Piazzesi, 2005; Yogo, 2006). Firms with limited external financing flexibility must rely more heavily on internal cash flows for investment, creating a tight link between their production decisions and aggregate consumption conditions. This connection becomes especially pronounced under habit formation preferences, where consumption risk premiums vary with the surplus consumption ratio (Campbell and Cochrane, 1999). High AFR firms effectively serve as levered claims on aggregate consumption growth, with their value being particularly sensitive to shocks that threaten the representative agent's consumption habit, thereby commanding higher equilibrium returns.

We find strong evidence that Asset Financing Restraint predicts cross-sectional variation in expected returns. The value-weighted long/short AFR strategy generates an annualized gross Sharpe ratio of 0.52, placing it in the top 9% of documented anomalies in the factor zoo. The strategy delivers a monthly average excess return of 26 basis points with a t-statistic of 3.73, demonstrating robust statistical significance. The economic magnitude remains substantial after controlling for established risk factors, with the monthly alpha relative to the Fama-French six-factor model equaling 22 basis points with a t-statistic of 3.05.

The predictive power of AFR persists across different portfolio construction methodologies and is not confined to small stocks. Among the largest quintile of stocks by market capitalization, the AFR strategy achieves a monthly average return of 36 basis points with a t-statistic of 4.09, with six-factor alphas of 28 basis points remaining highly significant. After accounting for transaction costs using high-frequency effective spreads, the value-weighted strategy maintains a net Sharpe ratio of 0.40, ranking in the top 5% of anomalies, and continues to expand the mean-variance efficient frontier beyond the Fama-French six-factor model with a generalized net alpha of 16 basis points per month.

Controlling for the six most closely related anomalies from the factor zoo—including Net debt financing, Net external financing, and Asset growth—the AFR strategy retains significant explanatory power with a monthly alpha of 18 basis points and a t-statistic of 2.57. The strategy loads positively on the CMA (conservative minus aggressive) factor with a coefficient of 0.28 and t-statistic of 5.87, consistent with the interpretation that high AFR firms maintain more conservative financing policies. These results demonstrate that AFR captures a distinct dimension of cross-sectional return variation related to firms’ external financing decisions and their implications for consumption risk exposure.

Our study contributes to the consumption-based asset pricing literature by identifying asset financing restraint as a novel characteristic that captures firms’ differential exposure to aggregate consumption risk. We extend the work of (Zhang, 2005) and (Li et al., 2009) on the asset pricing implications of financial constraints by demonstrating that the restraint in external financing relative to the asset base, rather than absolute financing levels, drives cross-sectional return predictability. Unlike (Cooper et al., 2008) who focus on total asset growth, we isolate the external financing component and show it contains distinct information about expected returns even after controlling for related growth and financing measures.

Methodologically, we employ the comprehensive testing framework of (Novy-Marx and Velikov, 2023) to establish the robustness of AFR across multiple dimensions, including alternative portfolio constructions, transaction costs, and controls for the entire factor zoo. This rigorous approach addresses concerns raised by (Harvey et al., 2016) about data mining and multiple testing in the anomalies literature. Our finding that AFR maintains significant predictive power after controlling for 209 other documented anomalies and exhibits a gross monthly alpha of 18 basis points relative to twelve factors (six Fama-French factors plus six closely related strategies) with a t-statistic of 2.57 demonstrates its incremental contribution to our understanding of the cross-section.

The economic significance of our findings extends beyond statistical robustness, offering new insights into how corporate financing decisions interact with macroeconomic risks to determine equilibrium asset prices. The strong performance of AFR among large-cap stocks, where it generates a monthly return of 36 basis points, suggests that financing constraints and their associated consumption risk exposures remain relevant even for firms with substantial market capitalizations. These results support consumption-based models that emphasize the role of financial frictions in amplifying firms' exposure to aggregate risks, particularly during economic downturns when both consumption growth is low and external financing becomes scarce (Gomes and Schmid, 2021).

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning multiple decades of annual observations. We focus on non-financial firms

and exclude utilities following standard practice in the asset pricing literature. Asset Financing Restraint signal relies on two key accounting variables. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new external financing raised through bond markets and term loans. Property, plant, and equipment gross (PPEGT) measures the total historical cost of tangible fixed assets before accumulated depreciation, representing the firm’s cumulative investment in productive capacity and infrastructure. These variables provide insight into firms’ financing activities and their existing asset base. We construct the Asset Financing Restraint signal by calculating the year-over-year change in long-term debt issuance scaled by lagged gross PPE: $(DLTIS(t) - DLTIS(t-1)) / PPEGT(t-1)$. This measure captures the change in a firm’s debt financing activity relative to its existing tangible asset base. A positive value indicates accelerating debt issuance relative to the firm’s physical capital stock, while a negative value suggests declining reliance on long-term debt financing. We compute this signal using annual observations at each fiscal year-end and require non-missing values for both DLTIS in consecutive years and lagged PPEGT. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. The scaling by gross PPE normalizes the financing change across firms of different sizes and capital intensities, allowing for meaningful cross-sectional comparisons.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the AFR signal. Panel A plots the time-series of the mean, median, and interquartile range for AFR. On average, the cross-sectional mean (median) AFR is -0.47 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input AFR data. The signal’s interquartile range spans -0.18 to 0.18. Panel B of Figure 1 plots the time-series of

the coverage of the AFR signal for the CRSP universe. On average, the AFR signal is available for 5.83% of CRSP names, which on average make up 7.10% of total market capitalization.

4 Does AFR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on AFR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high AFR portfolio and sells the low AFR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short AFR strategy earns an average return of 0.26% per month with a t-statistic of 3.73. The annualized Sharpe ratio of the strategy is 0.52. The alphas range from 0.22% to 0.31% per month and have t-statistics exceeding 3.05 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.28, with a t-statistic of 5.87 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 474 stocks and an average market capitalization of at least \$1,675 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies

are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 18 bps/month with a t-statistics of 4.02. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -8-26bps/month. The lowest return, (-8 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.40. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the AFR trading strategy improves the achievable mean-variance efficient frontier spanned by the

factor models in twenty cases, and significantly expands the achievable frontier in eighteen cases.

Table 3 provides direct tests for the role size plays in the AFR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and AFR, as well as average returns and alphas for long/short trading AFR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the AFR strategy achieves an average return of 36 bps/month with a t-statistic of 4.09. Among these large cap stocks, the alphas for the AFR strategy relative to the five most common factor models range from 28 to 40 bps/month with t-statistics between 3.11 and 4.55.

5 How does AFR perform relative to the zoo?

Figure 2 puts the performance of AFR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the AFR strategy falls in the distribution. The AFR strategy’s gross (net) Sharpe ratio of 0.52 (0.40) is greater than 91% (95%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the AFR strategy (red line).² Ignoring trading costs, a \$1 invested in the AFR strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that

would have yielded \$3.35 which ranks the AFR strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the AFR strategy would have yielded \$2.04 which ranks the AFR strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the AFR relative to those. Panel A shows that the AFR strategy gross alphas fall between the 59 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The AFR strategy has a positive net generalized alpha for five out of the five factor models. In these cases AFR ranks between the 77 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does AFR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of AFR with 209 filtered anomaly

a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price AFR or at least to weaken the power AFR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of AFR conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{AFR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{AFR}AFR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{AFR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on AFR. Stocks are finally grouped into five AFR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted AFR trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on AFR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the AFR signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

2.25, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on AFR is 1.32.

Similarly, Table 5 reports results from spanning tests that regress returns to the AFR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the AFR strategy earns alphas that range from 18-20bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.61, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the AFR trading strategy achieves an alpha of 18bps/month with a t-statistic of 2.57.

7 Does AFR add relative to the whole zoo?

Finally, we can ask how much adding AFR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the AFR signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes AFR grows to \$1152.06.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which AFR is available.

8 Conclusion

This study provides robust evidence that Asset Financing Restraint (AFR) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that AFR-based trading strategies generate substantial risk-adjusted returns even after accounting for transaction costs and controlling for established risk factors and related anomalies.

The key findings of our research underscore the economic importance of AFR as an asset pricing signal. The value-weighted long/short strategy delivers an annualized net Sharpe ratio of 0.40, which remains economically significant after accounting for implementation costs. More importantly, the strategy’s ability to generate a statistically significant alpha of 18 basis points per month (t -statistic = 2.57) even after controlling for the Fama-French five factors, momentum, and six closely related financing and investment strategies highlights that AFR captures unique information about future returns not subsumed by existing factors. This finding suggests that constraints in asset financing represent a distinct dimension of firm characteristics that influences expected returns beyond traditional measures of external financing, debt issuance, and asset growth.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, AFR adds to our understanding of how financing frictions affect asset prices and provides further evidence that capital market imperfections play a crucial role in determining the cross-section of expected returns. For practitioners, the robust performance of AFR-based strategies, particularly after accounting for transaction costs, suggests potential opportunities for portfolio enhancement through the incorporation of asset financing constraints into investment processes.

However, several limitations of this study warrant acknowledgment. First, while

we follow the Novy-Marx and Velikov (2023) protocol to ensure robustness, the sample period and market conditions examined may not fully capture the strategy’s performance across all market cycles. Second, the implementation of AFR strategies in practice may face additional frictions beyond the transaction costs we model, including market impact costs for large-scale implementations and potential capacity constraints. Third, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question.

Future research should address several important extensions of this work. First, examining the performance of AFR in international equity markets would provide valuable insights into the global applicability of this signal. Second, investigating the underlying economic mechanisms that drive the AFR premium would enhance our theoretical understanding of why financing constraints predict returns. This could include examining the role of agency costs, information asymmetries, or real investment distortions. Third, exploring the time-variation in the AFR premium and its relationship to macroeconomic conditions, credit market cycles, and monetary policy regimes would provide insights into when the strategy performs best. Finally, investigating potential interactions between AFR and other firm characteristics, such as profitability, investment, or governance measures, could reveal important conditional relationships that enhance our understanding of how financing constraints interact with other determinants of expected returns.

In conclusion, this paper establishes Asset Financing Restraint as a robust and economically significant predictor of stock returns that provides incremental information beyond existing factors and related strategies. While limitations exist and further research is needed to fully understand the economic mechanisms at work, our findings contribute meaningfully to the asset pricing literature and offer practical insights for portfolio construction and risk management.

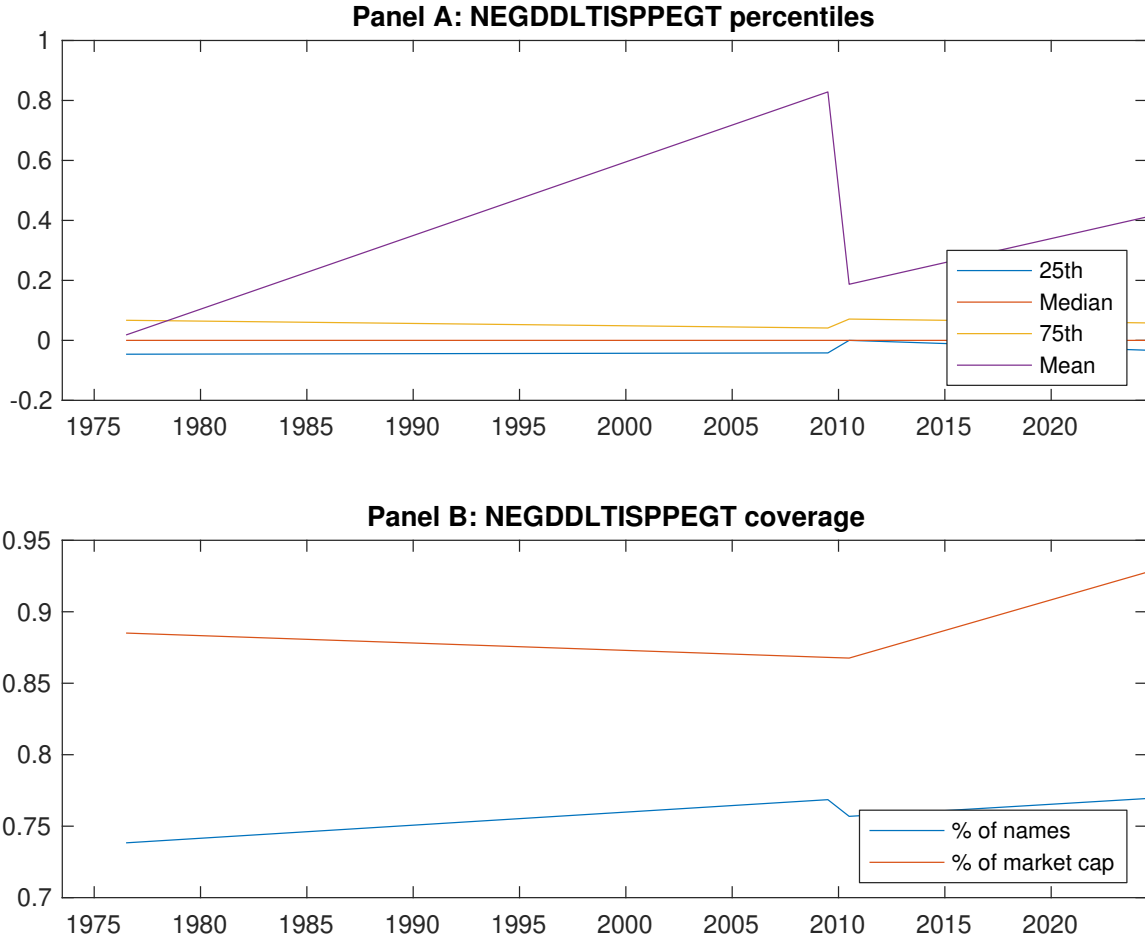


Figure 1: Times series of AFR percentiles and coverage.
This figure plots descriptive statistics for AFR. Panel A shows cross-sectional percentiles of AFR over the sample. Panel B plots the monthly coverage of AFR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on AFR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on AFR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.62 [2.84]	0.66 [3.77]	0.62 [3.22]	0.70 [4.01]	0.88 [4.30]	0.26 [3.73]
α_{CAPM}	-0.13 [-2.29]	0.07 [1.35]	-0.03 [-0.46]	0.11 [2.21]	0.18 [3.43]	0.31 [4.44]
α_{FF3}	-0.12 [-2.14]	0.05 [1.02]	0.03 [0.57]	0.11 [2.24]	0.20 [3.81]	0.31 [4.46]
α_{FF4}	-0.09 [-1.63]	0.05 [1.08]	0.08 [1.42]	0.07 [1.36]	0.18 [3.50]	0.27 [3.86]
α_{FF5}	-0.09 [-1.74]	-0.04 [-0.85]	0.06 [1.03]	0.03 [0.61]	0.14 [2.74]	0.24 [3.38]
α_{FF6}	-0.08 [-1.42]	-0.03 [-0.63]	0.09 [1.67]	0.00 [0.05]	0.14 [2.62]	0.22 [3.05]
Panel B: Fama and French (2018) 6-factor model loadings for AFR-sorted portfolios						
β_{MKT}	1.10 [86.47]	0.97 [86.31]	0.96 [73.15]	0.95 [82.62]	1.05 [86.53]	-0.05 [-2.80]
β_{SMB}	0.10 [5.18]	-0.11 [-6.52]	-0.02 [-0.82]	-0.05 [-2.66]	0.15 [8.16]	0.05 [2.05]
β_{HML}	0.00 [0.00]	0.02 [0.99]	-0.16 [-6.28]	-0.05 [-2.32]	-0.13 [-5.56]	-0.13 [-4.14]
β_{RMW}	0.05 [2.16]	0.15 [6.81]	-0.02 [-0.73]	0.09 [3.91]	0.07 [2.88]	0.01 [0.47]
β_{CMA}	-0.15 [-4.06]	0.15 [4.67]	-0.05 [-1.23]	0.18 [5.34]	0.13 [3.65]	0.28 [5.87]
β_{UMD}	-0.03 [-2.10]	-0.02 [-1.47]	-0.06 [-4.37]	0.04 [3.89]	0.01 [0.63]	0.03 [2.10]
Panel C: Average number of firms (n) and market capitalization (me)						
n	668	474	995	516	644	
me (\$10 ⁶)	1675	2634	2004	2612	1678	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the AFR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.26 [3.73]	0.31 [4.44]	0.31 [4.46]	0.27 [3.86]	0.24 [3.38]	0.22 [3.05]
Quintile	NYSE	EW	0.18 [4.02]	0.20 [4.65]	0.19 [4.35]	0.18 [4.16]	0.18 [4.06]	0.18 [4.00]
Quintile	Name	VW	0.27 [3.86]	0.31 [4.40]	0.32 [4.49]	0.29 [4.09]	0.25 [3.57]	0.24 [3.41]
Quintile	Cap	VW	0.26 [4.00]	0.31 [4.78]	0.31 [4.86]	0.27 [4.22]	0.23 [3.57]	0.21 [3.23]
Decile	NYSE	VW	0.33 [3.52]	0.39 [4.11]	0.37 [3.89]	0.34 [3.52]	0.23 [2.42]	0.22 [2.32]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.20 [2.88]	0.24 [3.46]	0.24 [3.45]	0.22 [3.14]	0.18 [2.56]	0.16 [2.32]
Quintile	NYSE	EW	-0.08 [-1.40]					
Quintile	Name	VW	0.21 [2.98]	0.24 [3.40]	0.24 [3.44]	0.23 [3.25]	0.19 [2.71]	0.18 [2.55]
Quintile	Cap	VW	0.21 [3.18]	0.25 [3.96]	0.26 [4.00]	0.24 [3.68]	0.19 [2.93]	0.17 [2.69]
Decile	NYSE	VW	0.26 [2.74]	0.32 [3.34]	0.30 [3.12]	0.28 [2.95]	0.18 [1.91]	0.17 [1.80]

Table 3: Conditional sort on size and AFR

This table presents results for conditional double sorts on size and AFR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on AFR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high AFR and short stocks with low AFR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	AFR Quintiles					AFR Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.67 [2.28]	0.87 [3.29]	0.97 [3.50]	0.92 [3.40]	0.84 [2.78]	0.16 [1.61]	0.20 [1.97]	0.17 [1.69]	0.14 [1.30]	0.13 [1.25]	0.11 [1.04]
	(2)	0.75 [2.75]	0.88 [3.55]	0.85 [3.34]	0.91 [3.74]	0.85 [3.29]	0.10 [1.18]	0.13 [1.65]	0.10 [1.32]	0.12 [1.50]	0.06 [0.80]	0.08 [1.00]
	(3)	0.83 [3.22]	0.78 [3.56]	0.83 [3.47]	0.80 [3.66]	0.94 [3.88]	0.11 [1.43]	0.16 [1.98]	0.15 [1.93]	0.11 [1.39]	0.16 [1.95]	0.13 [1.56]
	(4)	0.76 [3.25]	0.81 [3.85]	0.82 [3.71]	0.77 [3.77]	0.87 [3.92]	0.11 [1.49]	0.15 [1.94]	0.13 [1.66]	0.10 [1.25]	0.10 [1.31]	0.08 [1.04]
	(5)	0.53 [2.60]	0.63 [3.56]	0.58 [3.12]	0.58 [3.24]	0.89 [4.57]	0.36 [4.09]	0.40 [4.48]	0.40 [4.55]	0.34 [3.79]	0.32 [3.58]	0.28 [3.11]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	AFR Quintiles					AFR Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	367	368	369	368	364	33	30	29	29	32	
	(2)	101	101	101	102	101	55	56	55	56	55	
	(3)	72	73	72	73	73	100	100	96	97	100	
	(4)	61	62	62	62	61	219	225	218	222	217	
(5)	57	57	57	57	57	1394	2002	1667	2045	1472		

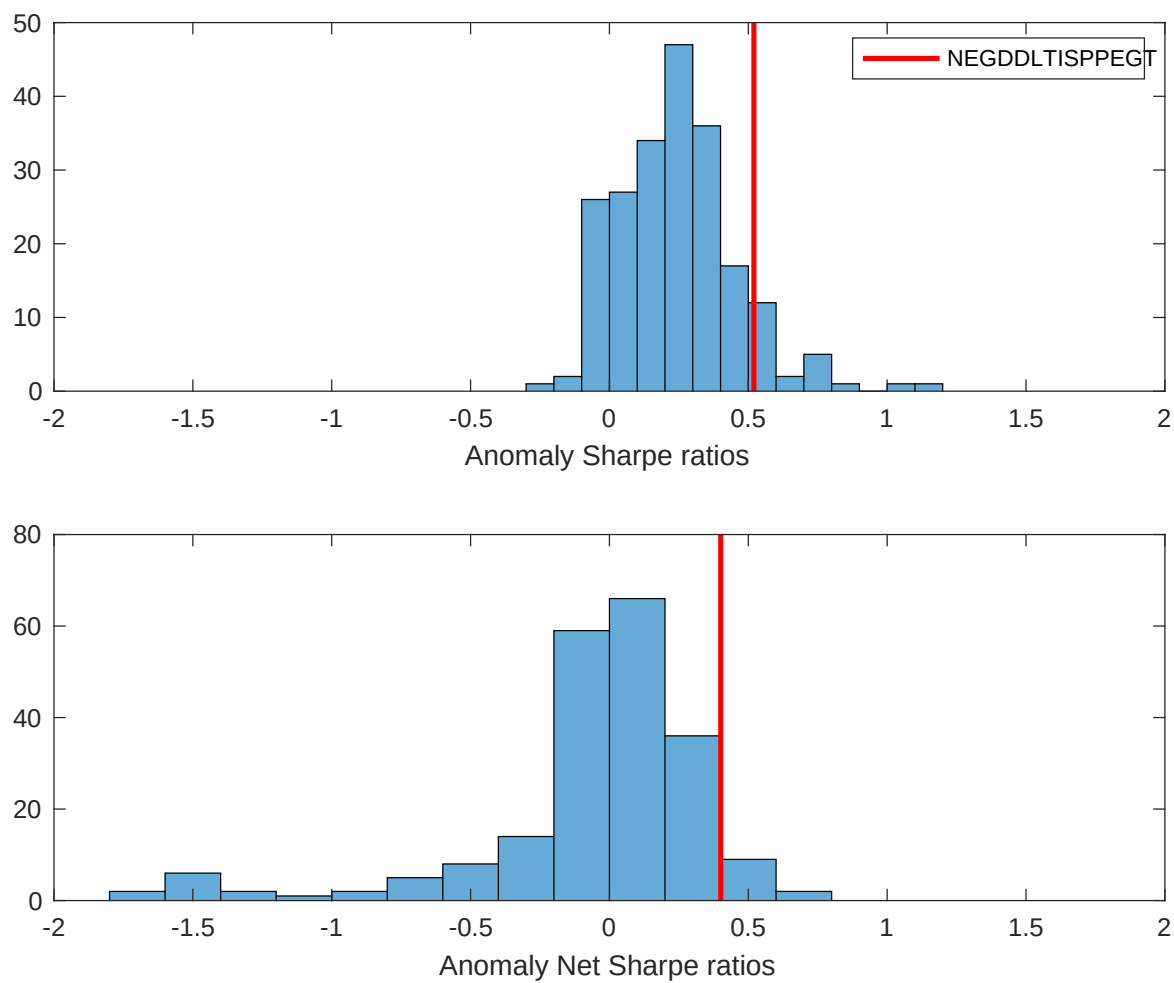


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the AFR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

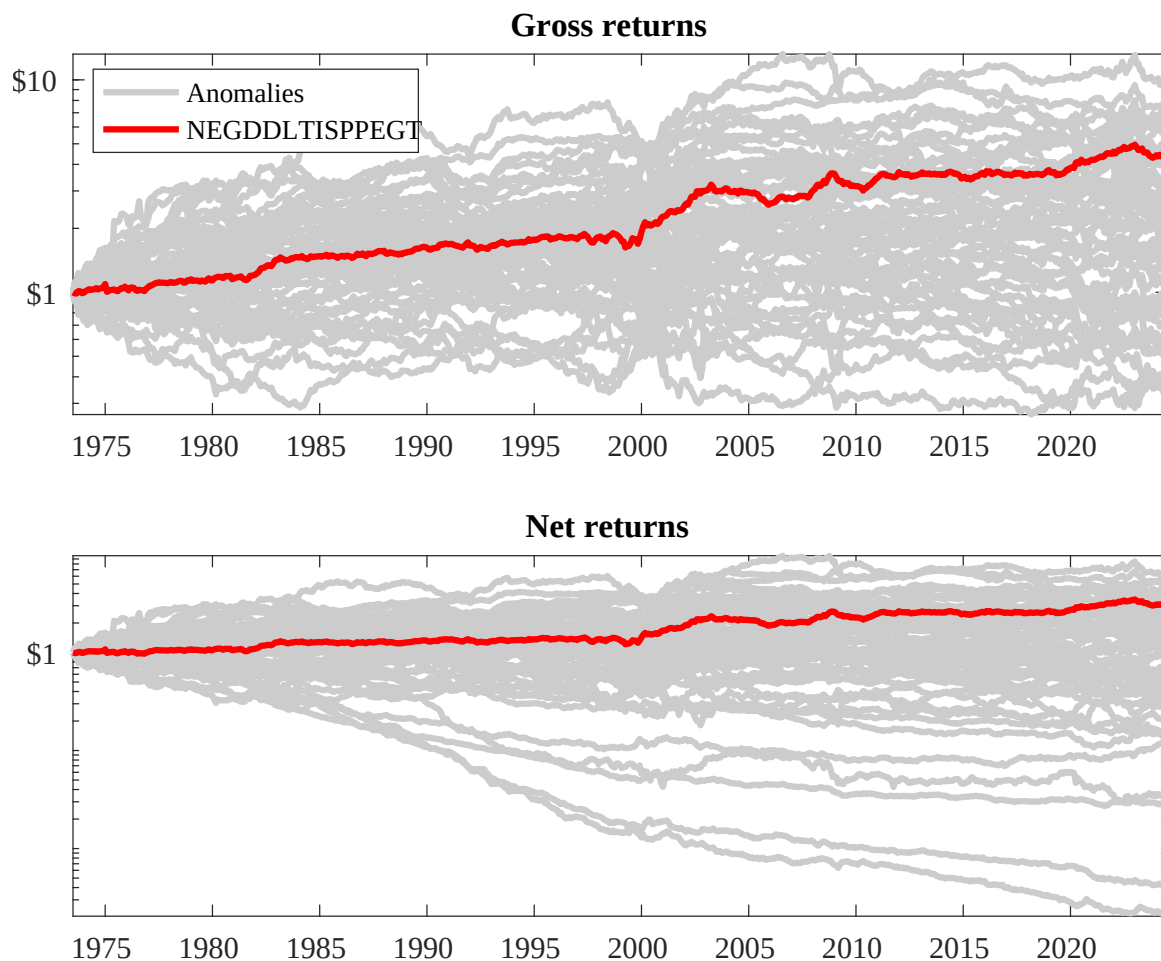
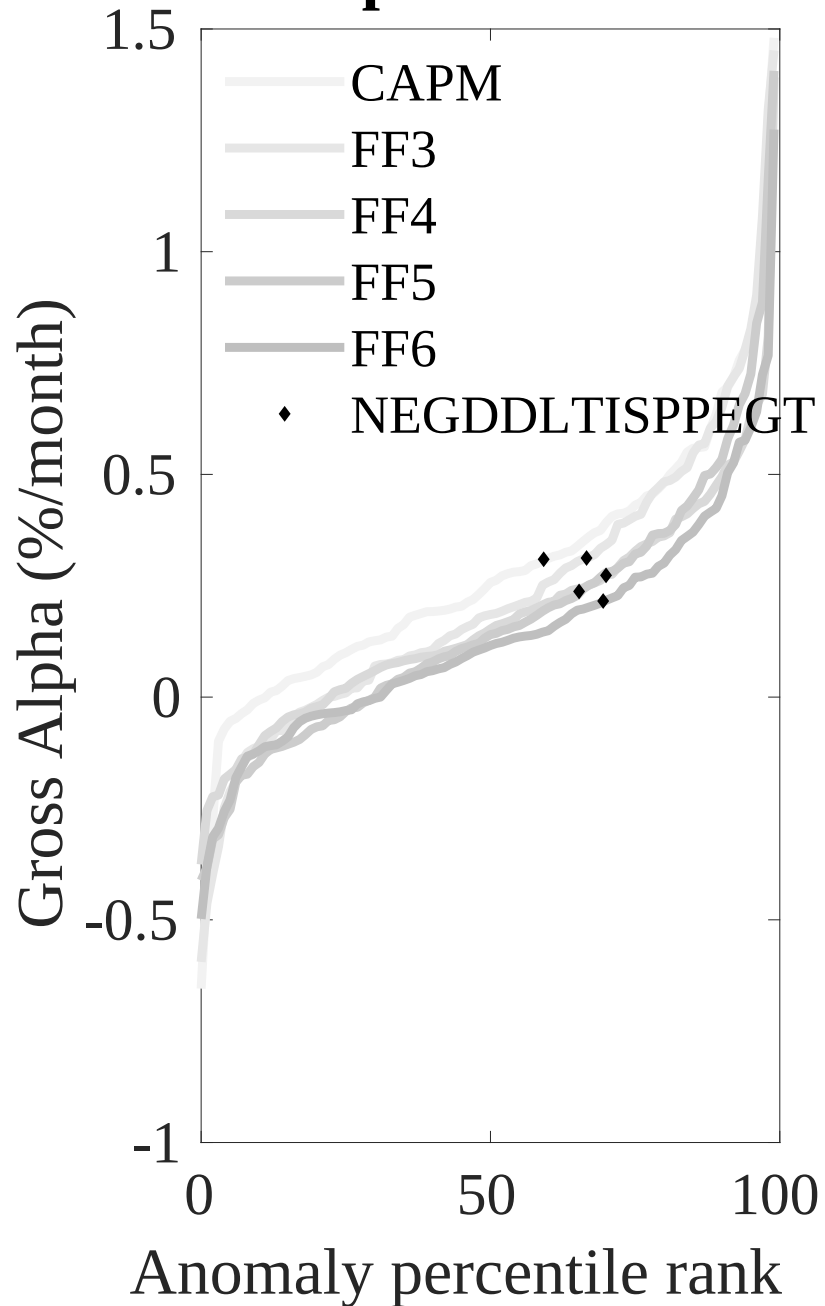


Figure 3: Dollar invested.

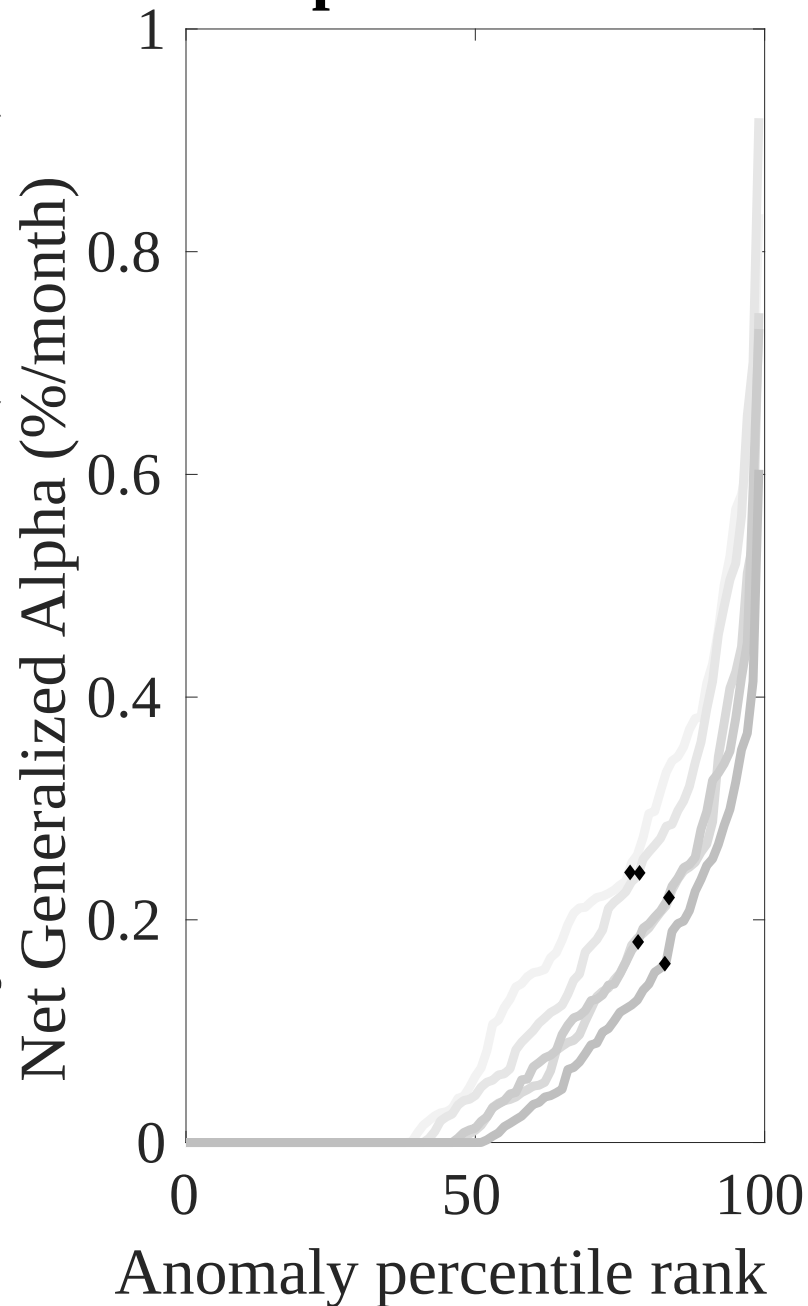
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the AFR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



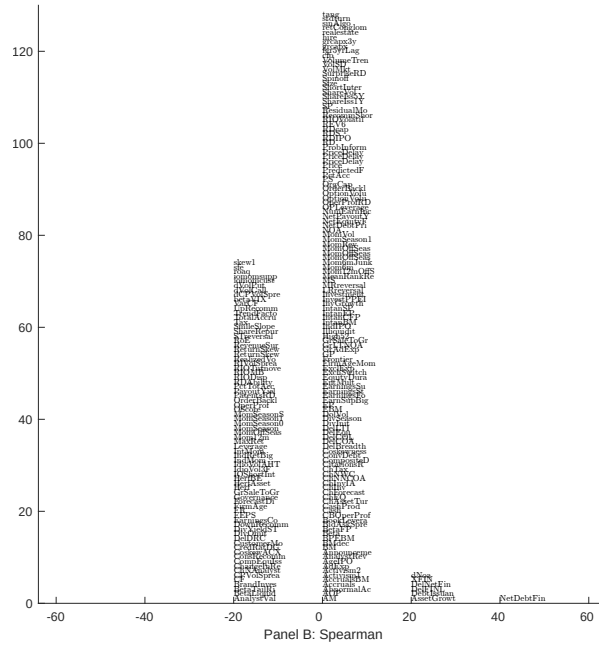
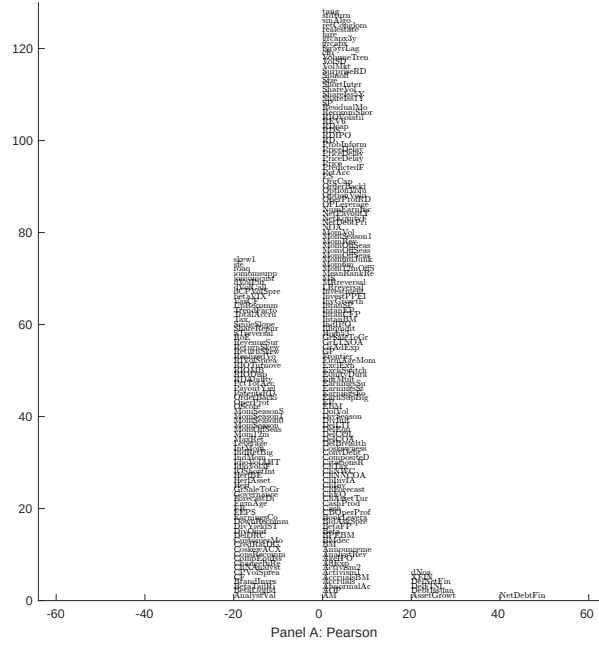


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with AFR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

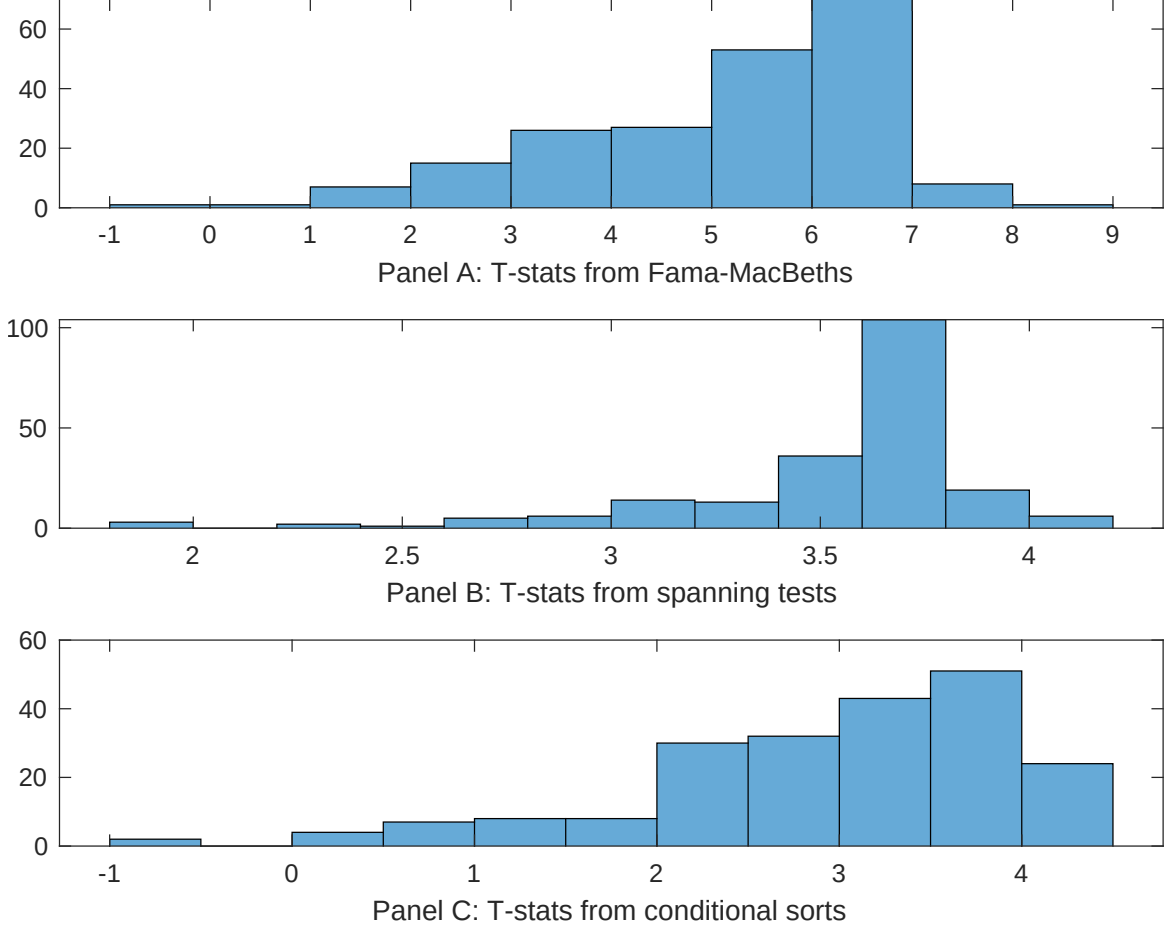


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of AFR conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{AFR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{AFR} AFR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{AFR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on AFR. Stocks are finally grouped into five AFR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted AFR trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on AFR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{AFR} AFR_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net debt financing, Net external financing, Change in financial liabilities, Asset growth, Inventory Growth, change in ppe and inv/assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.35]	0.14 [5.72]	0.13 [5.38]	0.14 [5.79]	0.14 [5.31]	0.14 [5.76]	0.15 [5.75]
AFR	0.88 [3.06]	0.10 [4.00]	0.75 [2.94]	0.64 [2.25]	0.15 [5.11]	0.95 [3.51]	0.42 [1.32]
Anomaly 1	0.19 [8.53]						0.91 [1.34]
Anomaly 2		0.19 [6.30]					0.11 [1.82]
Anomaly 3			0.17 [9.16]				-0.79 [-1.65]
Anomaly 4				0.10 [8.27]			0.51 [2.35]
Anomaly 5					0.41 [6.94]		0.25 [0.42]
Anomaly 6						0.17 [7.59]	0.86 [3.10]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	1	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the AFR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{AFR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net debt financing, Net external financing, Change in financial liabilities, Asset growth, Inventory Growth, change in ppe and inv/assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.19 [2.73]	0.18 [2.61]	0.19 [2.74]	0.20 [2.85]	0.20 [2.93]	0.20 [2.89]	0.18 [2.57]
Anomaly 1	15.76 [4.15]						11.45 [2.21]
Anomaly 2		17.80 [5.16]					14.30 [3.77]
Anomaly 3			9.47 [2.39]				-0.24 [-0.05]
Anomaly 4				9.79 [2.41]			3.87 [0.92]
Anomaly 5					4.97 [1.82]		3.92 [1.37]
Anomaly 6						4.40 [1.39]	-1.83 [-0.55]
mkt	-4.23 [-2.65]	-1.81 [-1.09]	-4.04 [-2.51]	-4.26 [-2.64]	-4.35 [-2.70]	-4.32 [-2.67]	-2.30 [-1.38]
smb	4.04 [1.66]	11.15 [4.22]	4.46 [1.81]	4.72 [1.92]	5.98 [2.43]	5.45 [2.23]	9.17 [3.26]
hml	-11.92 [-3.87]	-10.59 [-3.44]	-11.92 [-3.83]	-13.02 [-4.18]	-12.71 [-4.08]	-13.02 [-4.15]	-10.67 [-3.42]
rmw	-0.10 [-0.03]	-9.01 [-2.42]	0.78 [0.25]	1.67 [0.53]	2.35 [0.74]	1.71 [0.54]	-7.61 [-1.99]
cma	24.20 [5.07]	16.69 [3.24]	25.11 [5.12]	16.44 [2.42]	23.86 [4.49]	24.79 [4.63]	9.40 [1.34]
umd	2.63 [1.64]	3.52 [2.24]	2.83 [1.73]	3.85 [2.40]	3.14 [1.93]	3.64 [2.27]	2.50 [1.53]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	13	14	11	11	11	10	15

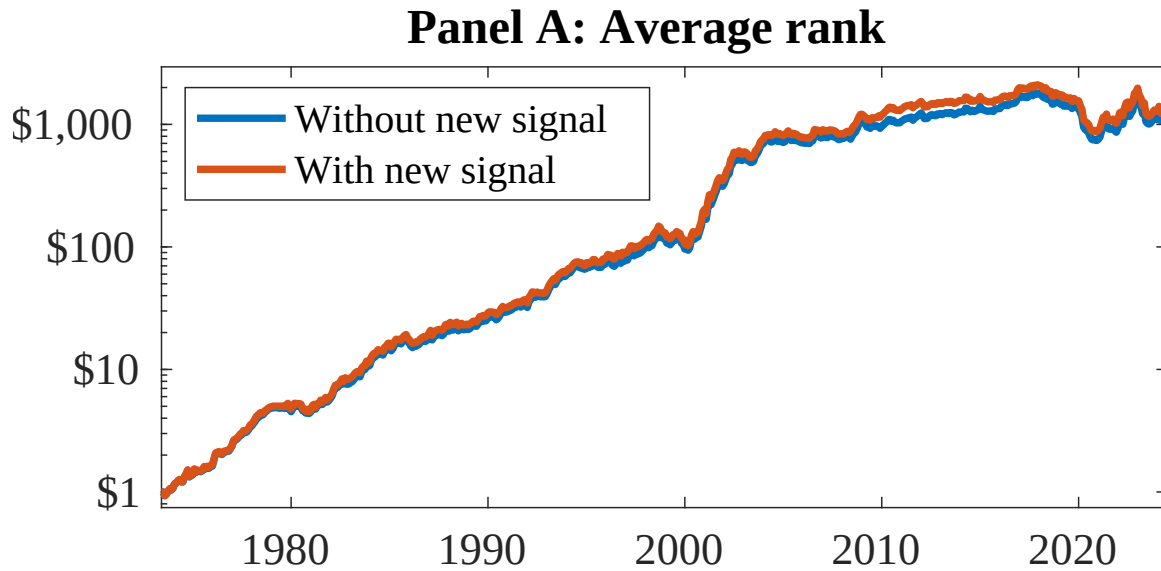


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as AFR. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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